CPU

CPU

- CPU, also called microprocessor or simply processor, is the brain of PC.
- It performs the system's data processing activity like calculation, comparison, etc.
- It is the most expensive single component in the system, costing four or more times than the motherboard it plugs into.
- The first CPU was created in 1971 by Intel. It is called the 4004.
- Brain of PC: it directly or indirectly controls the functions and activities of your computer.
- It transmits information between devices and tells the devices what to do.
- This is done by sending control information.

- Major producers of CPU are:
    - Intel
    - AMD
    - Cyrix
    - Sparc

Front Side Bus

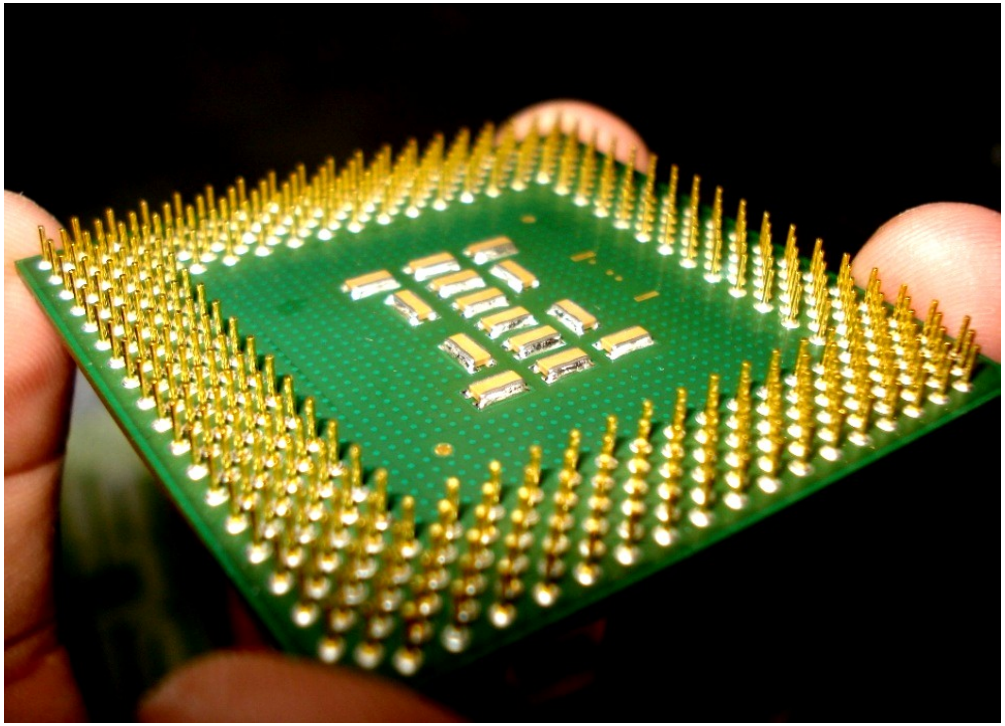
- Front side bus or system bus is a data bus that carries all information between CPU and all other devices within the system: RAM, AGP Video Card, PCI Expansion Cards, hard disk, etc.
- It is also called data bus.
- All devices are connected to the bus. Because of this, any information placed on the bus is available to all devices connected to the computer.

Processor Sockets and Slots

Socket: CPU socket is the connector that link CPU to motherboard particularly those compatible with Intel x86 architecture.

Most CPU today are built around PGA(Pin Grid Array) architecture where the **pins** in the underside of CPU are inserted into the socket. In contrast to this, several current and upcoming CPUs use LGA(Land Grid Array) where the **pins** are on the socket and they come in contact with the processor.

Processor Sockets and Slots…



- Fig CPU pins that fit to sockets

Processor Sockets and Slots…

- **Slot**: in this case, processors are cartridge shaped and fix into slot that looks like expansion slot.
- The CPU looks like card.

Processor Sockets and Slots…

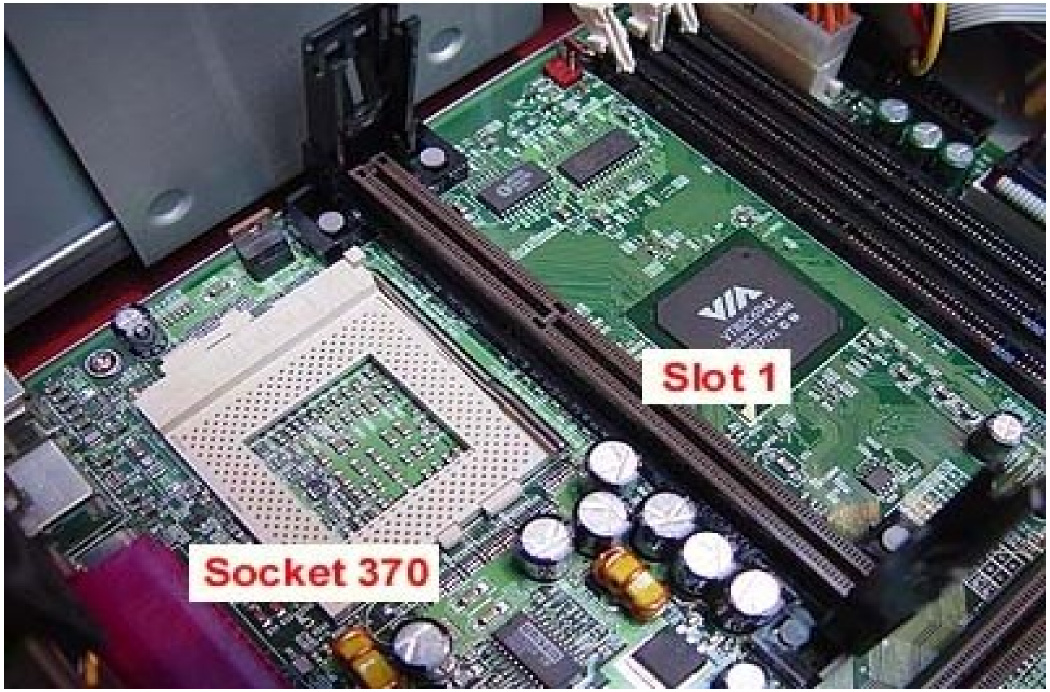


Fig SEC CPU and Slot

Both slot and socket are used to connect CPU to motherboard.

Processor Sockets and Slots…

Socket number	Pin number	Pin layout	Supported processors
Socket 1	169	PGA	486
Socket 2	238	PGA	486
Socket 3	237	PGA	486
Socket 4	273	PGA	Pentium
Socket 5	320	SPGA	Pentium, AMD K5
Socket 6	235	PGA	486
Socket 7	321	SPGA	Pentium, AMD K5/K6, cyrix MI/II
Socket 8	387	SPGA	Pentium Pro
Socket 370	370	SPGA	Pentium III, Celeron
Slot A	242	Slot	AMD Athlon
Socket A	462	PGA	AMD Athlon, AMD Duron
Slot 1	242	Slot	PII/PIII, Celeron
Slot 2	330	Slot	PII/PIII Xeon
Socket 423	423	SGPA	Pentium IV
Socket 478	478	PGA	P IV, Celeron
Socket T (LGA 775)	775	LGA	P IV, Celeron D

Socket T: uses Land Grid Array and is the latest socket type.
PGA: Pin Grid Array
SPGA: Staggered Pin Grid Array

Processor Sockets and Slots…

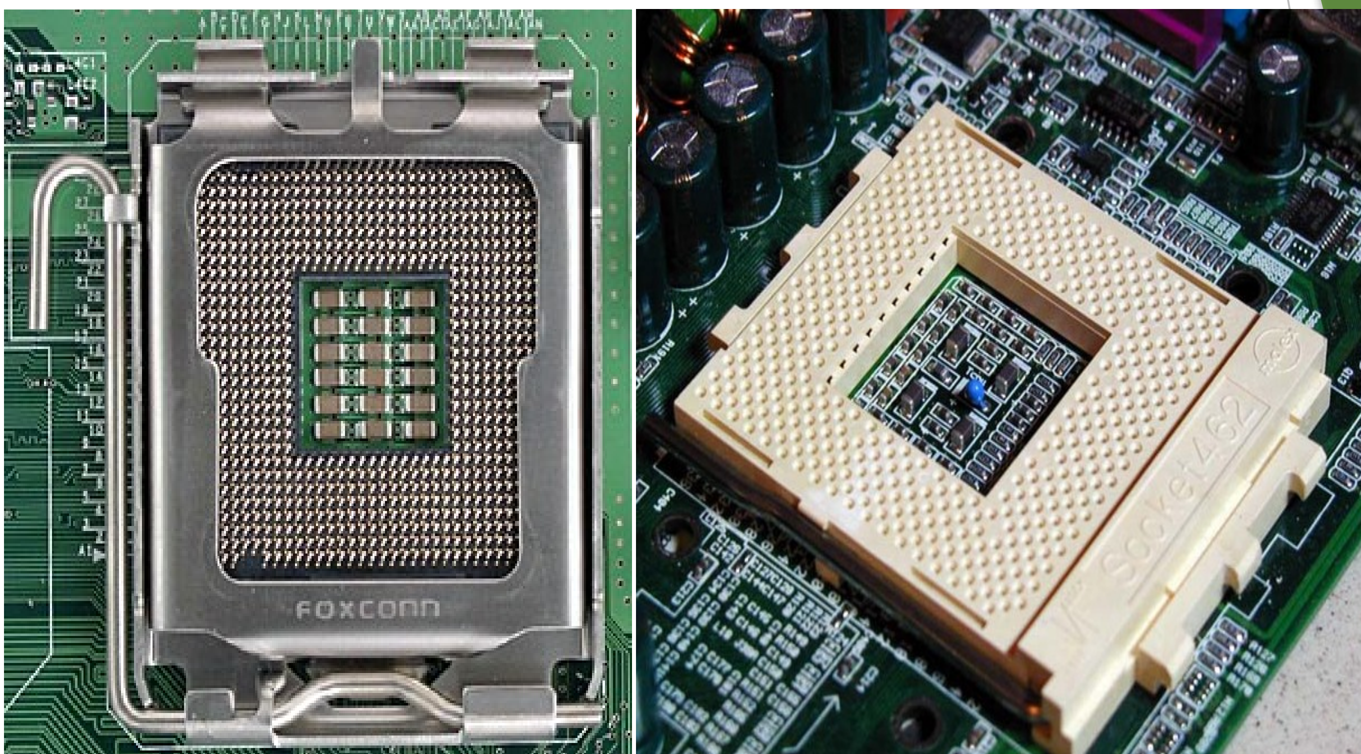
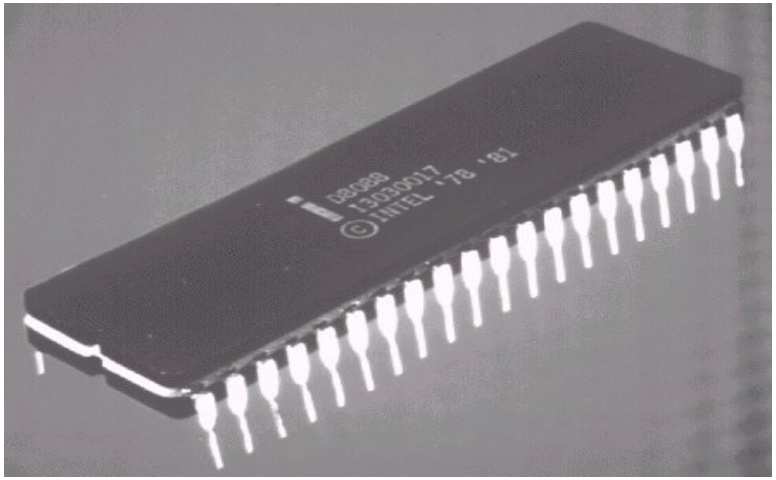


Fig CPU sockets (LGA, and SPGA)

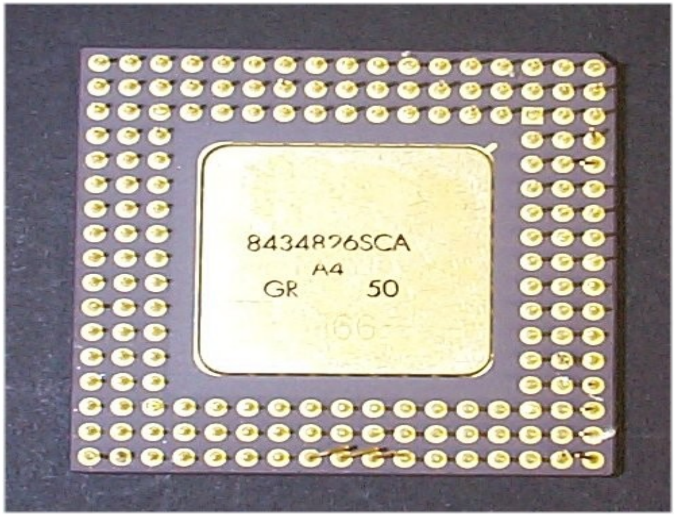
CPU Socket Types

1) Dual Inline Pin Package (DIPP): it has a set of pins that protrude in **two** sides of the CPU. Used in 8086, 8088, 80286 CPUs.



CPU Socket Types…

2) Pin Grid Array (PGA): the pins form straight line.



CPU Socket Types…

3) Staggered Pin Grid Array (SPGA): the pins form diagonal line.

CPU Socket Types…

4) Single Edge Contact (SEC): no pins and it uses slot.

CPU Cooling

- Specialized cooling system became serious starting from 80486. Earlier chips ran at a low speed and contained relatively few transistors. Because of this they needed no specialized cooling. But increase in CPU speed increased the heat generated necessitating cooling system.
- CPU cooling uses heat sinks. Heat sinks conduct heat from CPU to heat sink and then radiate it to air.
- Good cooling depends on the transfer of heat between the CPU and heat sink.
- The heat sink and CPU should contact each other to the maximum surface area possible. This allows heat to flow easily.

CPU Cooling…

There are two cooling mechanisms:

1. Passive Heat sink:-

- Have no **moving** parts.
- They are made up of aluminum.
- They cool the CPU by using thermal conduction and radiation.
- The heat sink draws heat from the CPU and air flowing through the heat sink cools the heat sink itself.
- The heat sink is clipped or glued to the processor.
- Thermal compound/grease is used to glue heat sink and CPU together so that they contact each other fully and there is no gap in between.

CPU Cooling…

2. Active Heat Sink

- Active heat sink adds a small **fan** that blows directly onto the heat sink metal to ensured direct air flow.
- An active heat sink cools better than passive heat sink by forcing air flow. Unfortunately, the fans have short life span and they are the first thing to fail in most PCs.

CPU Cooling…

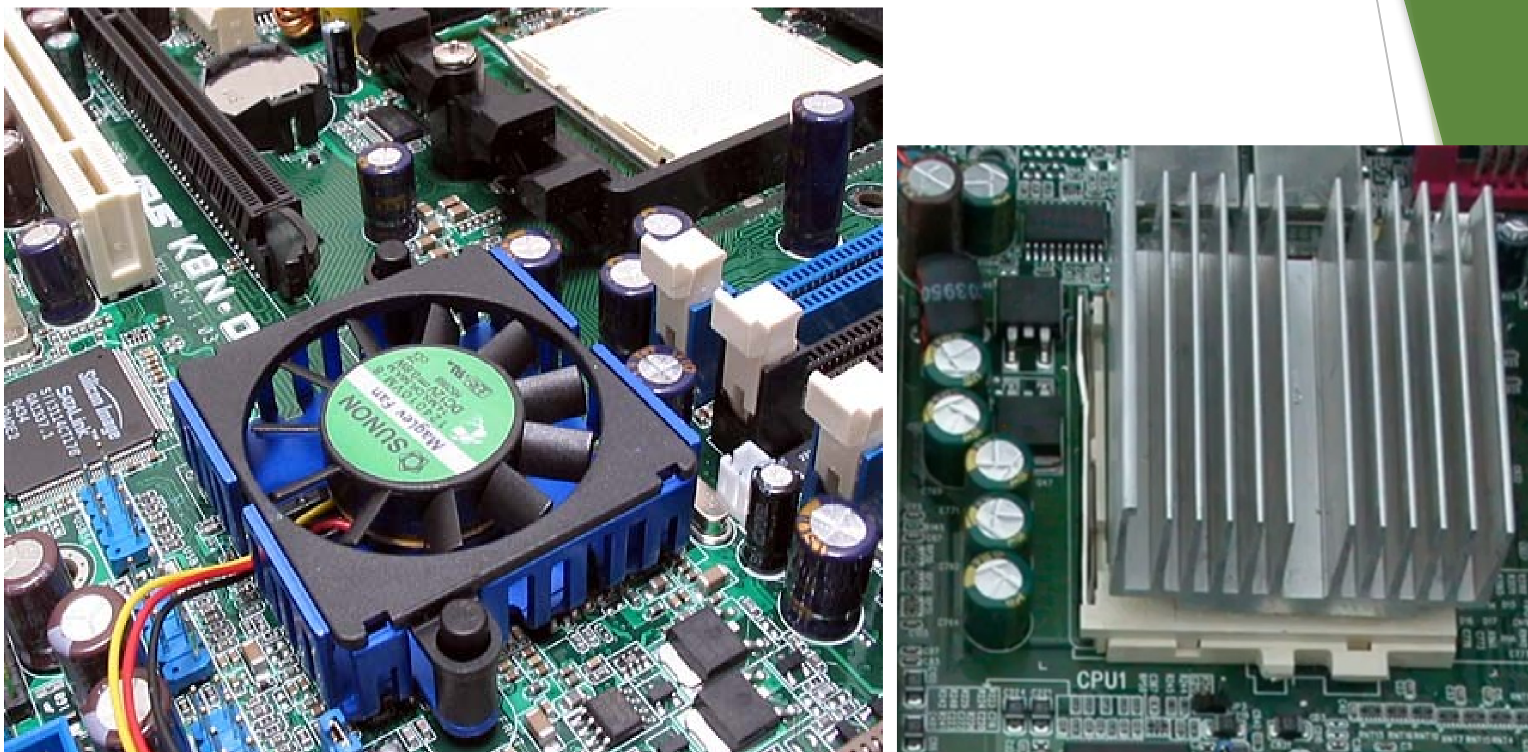


Fig CPU fan and heat sink

CPU Cooling…

Liquid Cooling

- On high end systems, liquid cooling is used
- Common in Mainframe computers
- It has reservoir, pump, and fan

Liquid cooling has three main parts:

- **Reservoir** where the liquid is stored
- **Pump** that pushes the liquid through pump to the CPU or other chipsets that should be cooled
- **Radiator** (fan) that cools the liquid

CPU Cooling…

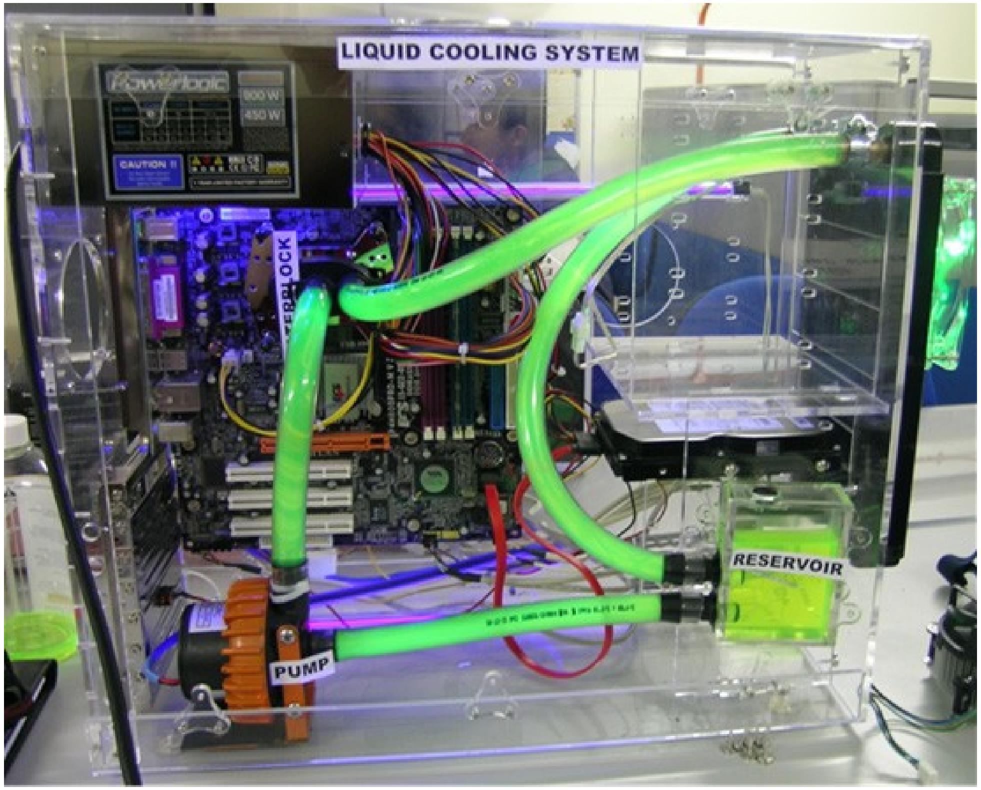


Fig liquid cooling system

CPU Cooling…

Overheated CPU causes:

- System crash
- Random reboot
- Memory errors
- Disk problems
- Application errors

Replacing and Upgrading CPU

Upgrade: installing another device which is better than the previous one in terms of performance.

- Poor upgrade may lead to total failure.

Replace: substitute with equivalent device in case the previous one failed.

- One thing to consider when we replace/upgrade CPU is the CPU socket on your motherboard.

Three most common sockets are:

- Low Insertion Force(LIF)
- Zero Insertion Force(ZIF)
- Single Edge Contact(SEC)

Replacing and Upgrading CPU…

LIF

- Removing an old CPU requires force. There are special tools that are designed for this.
- Use these tools:
  - Flat head screw driver
  - Chip puller

ZIF

- In ZIF, no force needed to remove or insert CPU into its socket.
- It is the most popular socket for PCs.
- It is easy to remove and insert CPU.
- It is introduced in early 1990s as a means to provide user-friendly CPU upgrade.

Replacing and Upgrading CPU…

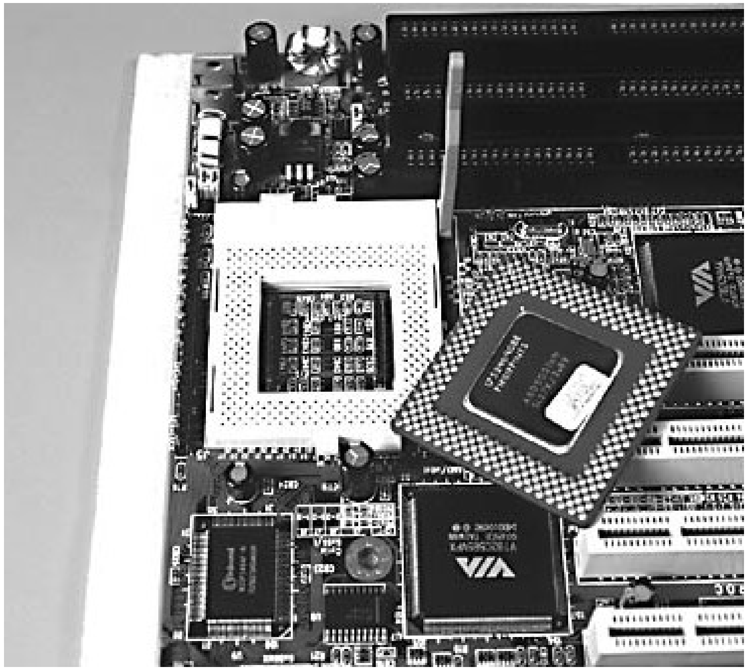


Fig ZIF socket and CPU

Replacing and Upgrading CPU…

- To **maximize** performance, you always **upgrade** to the fastest processor your particular board will support. This is determined by the type of socket on your motherboard. For example, if your motherboard has slot 1, you can't upgrade to Pentium IV which uses socket type connector instead of slot. Therefore, when you **buy** a new CPU for upgrade, select the type of CPU that **matches** with the socket on your motherboard.

Replacing and Upgrading CPU…

Upgrade Advice:

- Be careful when you handle CPU or any exposed IC. Static discharge could damage the chip.
- Take care not to bend any pins
- If you encounter any resistance, stop and figure out what is wrong. Do not use any force to insert the CPU.

Troubleshooting CPU Problems

- If the CPU does not work, the system gives beep sounds when turned on.
- The number of beeps varies from BIOS to BIOS. On AMI BIOS, **five** beeps indicate CPU problem.
  - Visit the **internet** for list of such errors.

Installing CPU:

- Turn off the PC and unplug the power cable
- disconnect external devices like mouse, keyboard, etc.
- follow appropriate ESD procedure
- remove the cover/case
- install CPU in its socket/slot
- install heat sink and fan (don’t forget fan power)